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Site-directed mutagenesis of the chemokine receptor CXCR6 suggests a novel paradigm for interactions with the ligand CXCL16

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Chemokine receptor CXCR6 mediates the chemotaxis and adhesion of leukocytes to soluble and membrane-anchored forms of CXCL16, and is an HIV-1 co-receptor. Here, we describe the effects of mutation of acidic extracellular CXCR6 residues on receptor function. Although most CXCR6 mutants examined were expressed at levels similar to wild-type (WT) CXCR6, an N-terminal E3Q mutant was poorly expressed, which may explain previously reported protective effects of a similar single nucleotide polymorphism, with respect to late-stage HIV-1 infection. In contrast to several other chemokine receptors, mutation of the CXCR6 N terminus and inhibition of post-translational modifications of this region were without effect on receptor function. Likewise, N-terminal extension of CXCL16 resulted in a protein with decent potency and efficacy in chemotaxis and not, as anticipated, a CXCR6 antagonist. D176N and E274Q CXCR6 mutants were unable to interact with soluble CXCL16, suggesting a critical role for D176 and E274 in ligand binding. Intriguingly, although unable to interact with soluble CXCL16, the E274Q mutant could promote robust adhesion to membrane-anchored CXCL16, suggesting that soluble and membrane-bound forms of CXCL16 possess distinct conformations. Collectively, our data suggest a novel paradigm for the CXCR6:CXCL16 interaction, a finding which may impact the discovery of small-molecule antagonists of CXCR6.

Key words: Chemokine · CXCR6 · G protein-coupled receptor · Molecular biology

Introduction

Recruitment of leukocytes to sites of inflammation is coordinated by the production of chemokines, small proteins which interact with specific chemoattractant receptors belonging to the superfamily of G protein-coupled receptors [1]. Although typically produced as soluble, secreted proteins, the chemokine CXCL16 also exists as a membrane-bound form, consisting of a chemokine domain, a mucin-type stalk, a single-pass transmembrane (TM) domain and a cytoplasmic tail [2, 3]. The mucin stalks serve to tether the chemokines to the cell membrane, while soluble versions of CXCL16 can be released by the action of membrane metalloproteinases [4, 5].

CXCL16 is expressed by APC, including subsets of $CD14^+$ monocytes/macrophages and $CD19^+$ B cells [3], and has a variety of functions. Membrane-bound, it serves as a scavenger receptor for phosphatidylserine and oxidised low-density lipoprotein [6], and facilitates the phagocytosis of bacteria [7]. Immunohistochemical analysis suggests that CXCL16 is expressed by lipid-laden macrophages in the intima of atherosclerotic plaques and by valvular and neocapillary endothelial cells in patients with infective endocarditis, suggesting a role in the pathogenesis of both diseases [8, 9]. Membrane-bound CXCL16 also facilitates the firm adhesion of cells expressing the G protein-coupled receptor

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CXCR6, such as activated T cells and NKT cells [10]. As a soluble chemokine, CXCL16 promotes chemotaxis *via* CXCR6, and serves as a chemoattractant for $CXCR6^{+}$ T cells [2, 3].

Current understanding of chemokine-chemokine receptor binding and receptor activation comes from extensive studies of several receptors and proposes a two-step model where the chemokine is first tethered to the receptor *via* a high-affinity interaction involving the N terminus of the receptor. This promotes a secondary, lower-affinity interaction involving the remaining extracellular domains [11, 12] and orientates the chemokine such that its own N terminus can enter an intrahelical pocket and induce an active receptor conformation and intracellular signalling [13, 14].

In this study, we describe a programme of point mutagenesis of CXCR6 to examine the CXCL16:CXCR6 interaction. We report the identification of key residues on CXCR6 involved in ligand binding and activation and demonstrate that the receptor can discriminate between different conformations of the soluble and membrane-bound forms of CXCL16.

Results

CXCR6 transfectants bind and migrate in response to soluble CXCL16

The murine pre-B cell line L1.2 was transiently transfected with a plasmid encoding CXCR6 tagged at the N terminus with the HA epitope. Cell surface expression was examined by flow cytometry and cells were subsequently used in chemotaxis and radioligand binding assays to determine their response to CXCL16 (Fig. 1). CXCR6 L1.2 cells were stained with anti-HA mAb, confirming the presence of CXCR6 at the cell surface (Fig. 1A), with little staining observed on naive cells (data not shown). Soluble CXCL16 was able to displace ^{125}I-CXCL16 with an EC_{50} of 37.7 nM (Fig. 1B, Table 1), while CXCR6 L1.2 cells migrated in response to CXCL16 with a typical bell-shaped profile (Fig. 1C) with 30 nM CXCL16 inducing optimal migration. Untransfected cells did not migrate to CXCL16 (data not shown), in agreement with the lack of cell surface staining.

The CHO-K1 cell line was stably transfected with a plasmid encoding CXCL16 cDNA and cell surface expression of membrane-bound chemokine was observed by flow cytometry using a CXCL16-specific antiserum (Fig. 1D). Since the action of endogenous metalloproteinases may facilitate the release of soluble CXCL16 into the tissue culture supernatant [4, 5], chemotaxis of CXCR6-transfected L1.2 cells to varying dilutions of culture media was assayed (Fig. 1E). CXCR6-expressing cells readily migrated to supernatant from CXCL16-expressing CHO cells, but not from naive CHO cells. This suggests that both the CXCR6 and CXCL16 produced in our model systems are functional. Interestingly, whilst the CXCL16 shed into the supernatant was functional in terms of inducing chemotaxis, it was unable to displace 0.1 nM ^{125}I-CXCL16 from the surface of CXCR6 transfectants (Fig. 1F) even at dilutions well in excess of those needed to induce a bell-shaped response in chemotaxis assays (Fig. 1E).

Probing the CXCR6:CXCL16 interaction by mutagenesis of the CXCR6 N terminus

Many chemokine receptors follow a two-step model of activation, where the chemokine is initially bound to the N terminus of the receptor through a high-affinity interaction, an event followed by receptor activation through a lower-affinity interaction with the extracellular loops (ECL) and TM domains. To study whether the CXCL16:CXCR6 interaction conforms to the two-step model, we initially generated chimeric constructs of CXCR1 and CXCR6, in which the N termini of CXCR1 and CXCR6 were exchanged, a method previously used to identify the domains involved in ligand binding and signalling for other receptors [11, 12]. Unfortunately, both chimeras were poorly expressed at the cell surface compared to their wild-type (WT) counterparts, and were non-functional in assays of chemotaxis, ^{125}I-CXCL16 binding, or adhesion (Table 1), leading us to examine the role of the CXCR6 N terminus by a program of site-directed mutagenesis.

Since basic residues of CXCL16 have previously been found to be important for binding to CXCR6 [15], we hypothesised that negatively charged residues on the receptor were likely candidates for electrostatic interactions with basic residues of the chemokine. Individual acidic residues within the N terminus were mutated to their uncharged counterpart, either glutamine or asparagine (Fig. 2). Coupled with this, we also explored the role of post-translational modifications (PTM) of this receptor region, since studies of CCR5, CX3CR1 and CXCR3 have shown that O-linked oligosaccharides and sulfated tyrosine residues within the N terminus play an important role in chemokine binding [16–19]. The impact of PTM on CXCR6 function was probed using both site-directed mutagenesis and pharmacological inhibitors of glycosylation and sulfation.

Mutation of acidic residues located within the N terminus, on an individual basis, had little effect on CXCR6 function as deduced by assays of ligand binding, chemotaxis or adhesion (Table 1). One notable exception was the residue E3, mutation of which resulted in a 70% loss in cell surface expression, compared with WT CXCR6. However, despite this lowered cell surface expression, the construct was still functional in all three assays.

Within the N terminus, two clusters of four acidic residues lie, one between residues 3 and 9 and one between residues 17 and 25 (Fig. 2). Three additional constructs were therefore generated, two in which either all four residues within each cluster were mutated to their neutral counterpart (cluster 1 and cluster 2) and one construct in which all eight acidic residues were mutated (cluster 1&2). Although cell surface expression was greatly reduced in constructs containing the E3Q mutant, all three cluster constructs were able to facilitate binding, migration and adhesion to soluble and membrane-bound chemokine (Table 1). This suggests that acidic residues within the CXCR6 N terminus are not critical for a productive interaction with CXCL16.

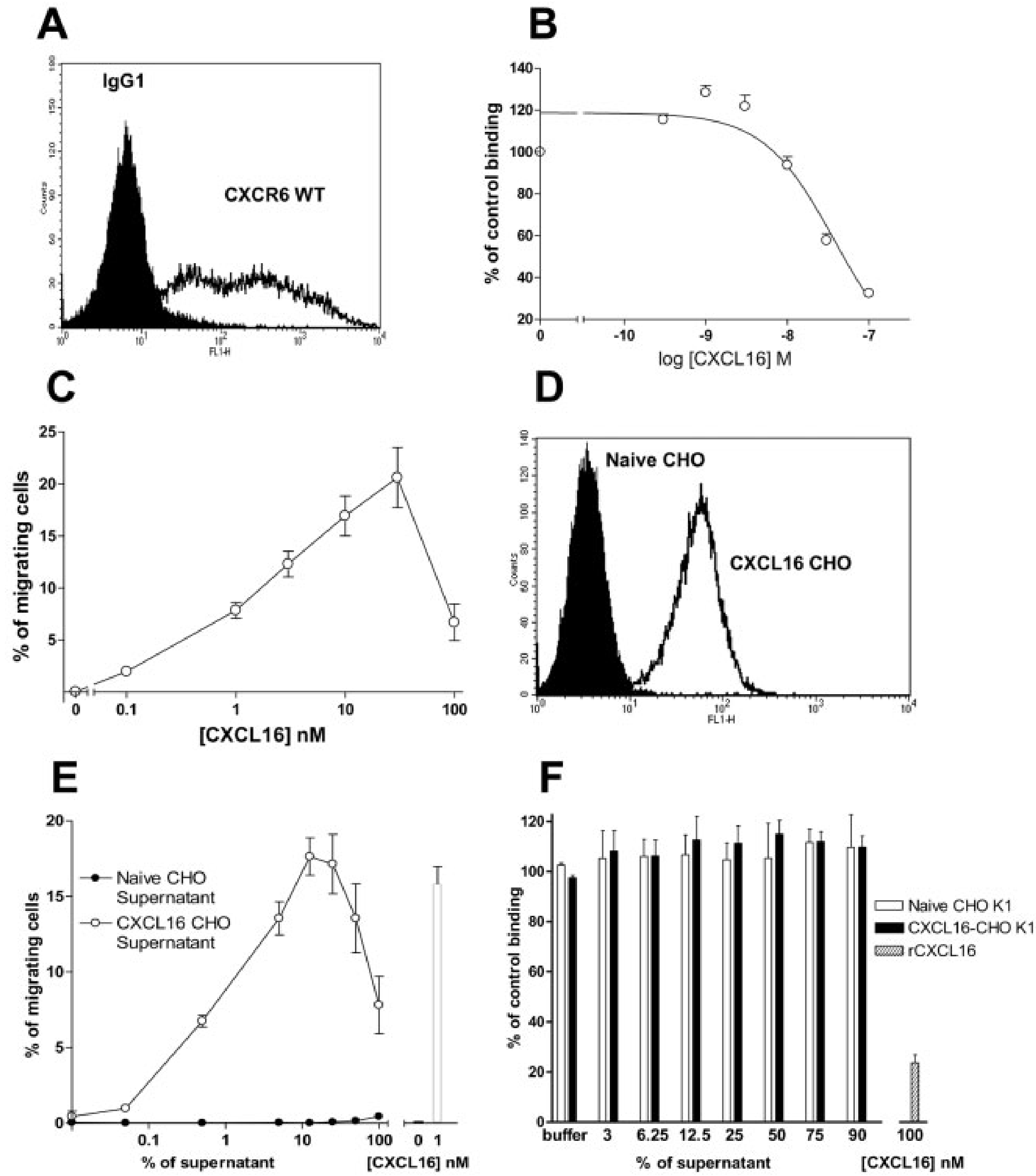

Figure 1. Functional expression of CXCR6 and CXCL16 in mammalian cells. (A) Typical cell surface expression of HA-CXCR6-transfected L1.2 cells as deduced by flow cytometry using an anti-HA mAb (open histogram) or isotype control (filled histogram). (B) Displacement of ^{125}I-CXCL16 by increasing concentrations of CXCL16 from WT CXCR6 L1.2 transfectants. (C) Chemotactic responses of WT CXCR6 L1.2 transfectants to increasing concentrations of CXCL16. (D) Representative cell surface expression of CXCL16 following the stable transfection of CHO-K1 cells. The open histogram represents staining with anti-CXCL16 Ab, whilst the filled curve is control goat IgG. (E) Chemotactic response of WT CXCR6 L1.2 transfectants to varying dilutions of supernatant from the CXCL16-CHO-K1 transfectants (open circles) and supernatant from naive CHO-K1 cells (filled circles). Responses to recombinant CXCL16 (white bars) are shown as a control. (F) Displacement of ^{125}I-CXCL16 by increasing concentrations of supernatant from naive CHO-K1 cells (open bars) or CXCL16-CHO-K1 transfectants (filled bars), or 100 nM recombinant CXCL16 (hatched bar).

To address PTM of the CXCR6 N terminus, we transfected L1.2 cells with plasmids encoding WT CXCR6 and subsequently cultured the cells with sodium chlorate (an inhibitor of sulfation) or tunicamycin (an inhibitor of N-linked glycosylation). Whilst sodium chlorate had little effect on cell surface expression, incubation with either tunicamycin or the vehicle control, DMSO, was seen to significantly reduce cell surface expression (Fig. 3A). None of the treatments was seen to abolish chemotactic responses to CXCL16, although the ligand induced a more potent, but less efficacious response from the tunicamycin-treated cells, correlating with the reduced expression (Fig. 3B). To address O-linked glycosylation, we transfected L1.2 cells with WT CXCR6 plasmid and cultured the cells with the inhibitor benzyl-2-acetamido-2-deoxy-α-D-galactopyranoside [20]. This had little effect upon CXCR6 cell surface expression, as did pre-incubation of WT CXCR6-expressing cells with sialidase, to remove terminal sialic

Table 1. Expression, chemokine binding, chemotaxis and adhesion of CXCR6 mutants[a)]

Mutation	Receptor/ mutant	% of WT surface expression[b)]	EC_{50} [nM] of chemotaxis	% of WT binding[b)]	IC_{50} [nM]	log IC_{50} [M] ± log SE	% of WT adhesion[b)]
WT	CXCR6	100	1.67 ± 0.27	100	37.7	–7.4 ± 0.1	100
N-terminal chimeras	Chimera 1	14.6 ± 0.4	ND	1.5 ± 1.2	ND	ND	0.0 ± 8.1
	Chimera 2	47.2 ± 5.0	ND	0.0 ± 1.4	ND	ND	0.0 ± 0.1
N-terminal acidic residues	E3Q	28.6 ± 5.6	1.1 ± 0.6	91.4 ± 40.0	173.2	–6.8 ± 0.6	94.4 ± 10.0
	D5N	95.2 ± 8.9	1.4 ± 0.4	92.4 ± 30.0	ND	ND	78.6 ± 7.5
	E8Q	76.7 ± 17.3	2.7 ± 0.3	82.8 ± 33.3	38.8	–7.4 ± 0.3	80.7 ± 24.9
	D9N	65.0 ± 15.0	3.9 ± 0.3	85.7 ± 30.0	47.4	–7.3 ± 0.3	84.9 ± 3.5
	D17N	91.8 ± 15.8	4.1 ± 0.3	83.5 ± 31.3	ND	ND	99.0 ± 13.8
	E21Q	76.8 ± 12.1	2.3 ± 0.3	106.1 ± 32.3	38.9	–7.4 ± 0.3	69.5 ± 34.6
	E22Q	80.0 ± 21.6	3.2 ± 0.3	76.1 ± 34.1	ND	ND	61.6 ± 25.6
	D25N	98.0 ± 18.8	1.6 ± 0.2	94.5 ± 10.4	67.0	–7.2 ± 0.3	68.2 ± 41.9
N-terminal cluster mutants	1	14.4 ± 3.7	0.2 ± 0.8	49.4 ± 21.5	22.6	–7.6 ± 0.3	112.5 ± 49.6
	2	76.6 ± 36.7	0.6 ± 0.5	82.6 ± 12.4	47.1	–7.3 ± 0.5	91.3 ± 10.2
	1&2	10.8 ± 5.0	0.1 ± 0.7	36.5 ± 14.8	14.8	–7.8 ± 0.2	87.7 ± 26.7
N-terminal PTM	Y6F	85.5 ± 21.9	0.9 ± 0.6	73.1 ± 11.8	34.8	–7.5 ± 0.3	80.2 ± 10.5
	Y10F	81.3 ± 21.4	0.9 ± 0.5	66.7 ± 4.2	40.2	–7.4 ± 0.3	111.1 ± 18.0
	Y6Y10F	80.6 ± 16.8	0.7 ± 0.6	60.0 ± 11.5	42.2	–7.4 ± 0.2	119.1 ± 33.8
	N16A	42.4 ± 11.0	3.5 ± 0.2	80.4 ± 10.0	9.8	–8.0 ± 0.2	70.5 ± 19.5
ECL acidic residues	E94Q	26.9 ± 10.5	ND	68.4 ± 26.3	9.1	–8.0 ± 0.2	ND
	D176N	56.7 ± 18.2	ND	2.7 ± 0.7	ND	ND	ND
	D184N	81.3 ± 27.3	2.7 ± 0.3	73.6 ± 24.2	29.3	–7.5 ± 0.2	41.6 ± 8.1
	E185Q	75.9 ± 29.1	1.2 ± 0.4	38.6 ± 21.2	37.2	–7.4 ± 0.3	82.0 ± 16.5
	E259Q	84.5 ± 28.5	1.2 ± 0.4	44.2 ± 17.7	36.7	–7.4 ± 0.3	85.5 ± 31.8
	E274Q	91.4 ± 39.6	ND	0.2 ± 1.6	ND	ND	58.3 ± 10.0

a) The mean values ± SEM from at least three experiments are shown; ND: not determined.
b) Surface expression, binding and adhesion are expressed as a comparison to values obtained for WT CXCR6 expression, binding and adhesion.

acid residues from oligosaccharides (Fig. 3C). Likewise, both treatments had little effect upon chemotactic responses to CXCL16 (Fig. 3D).

To corroborate the data obtained with the pharmacological inhibitors of PTM on N-linked glycosylation and sulfation, we mutated the sole asparagine and tyrosine residues within the CXCR6 N terminus to alanine and assessed the effects upon expression, chemotaxis, and CXCL16 adhesion and ligand binding (Fig. 3E–H, Table 1). In agreement with the tunicamycin data, mutation of N16 led to a significant decrease in cell surface expression of CXCR6, which correlated with reduced chemotaxis at the higher concentrations of CXCL16 compared with WT CXCR6 transfectants (Fig. 3F) and a reduced IC_{50} value in competition binding experiments (Table 1). No effect of the N16A mutation on adhesion to CXCL16-expressing cells or binding of soluble CXCL16 was apparent (Fig. 3G–H).

In keeping with the sodium chlorate data, mutation of Y6 or Y10, either alone or in combination, had no effect upon any of the

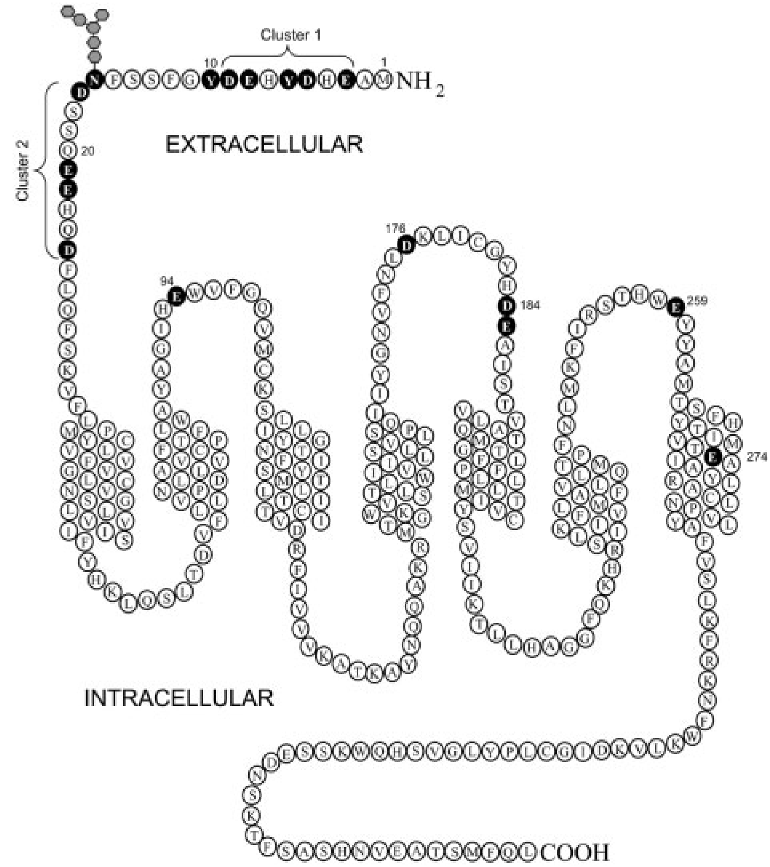

Figure 2. The likely topology of CXCR6. The sequence of CXCR6 was aligned with that of bovine rhodopsin, and putative TM helices determined by reference to the rhodopsin crystal structure [40]. Filled circles indicate amino acids which were investigated by point mutagenesis. The single N-linked glycosylation site at N16 is highlighted. Clusters of acidic residues within the N terminus are also highlighted.

CXCR6 activities examined (Fig. 3E–H). We therefore conclude that PTM within the CXCR6 N terminus are not critical for receptor function.

The N terminus of CXCL16 is unlikely to interact with an intrahelical pocket

The two-step model of chemokine receptor activation predicts the engagement of the chemokine N terminus with residues contained within an intrahelical receptor pocket, comprised of the TM helices. This pocket is of limited accessibility; indeed, extension of the N terminus of CCL5 by a single methionine generates a potent receptor antagonist [21]. To directly address the activation of CXCR6 by CXCL16, we generated a recombinant protein in *Escherichia coli*, in which the N terminus of mature CXCL16 was extended by 24 amino acids (Fig. 4A). This extension was comprised of a T7 gene 10 leader peptide, a 6-histidine tag and an enterokinase site, allowing efficient production and purification of recombinant protein. Following its production in *E. coli* and subsequent purification, the protein was analysed by non-reducing SDS-PAGE and compared to commercially available soluble CXCL16 chemokine domain (Fig. 4B). The commercially available CXCL16 ran at the anticipated molecular mass of 10 kDa, while the tagged version, as anticipated, ran slightly higher at approximately 12 kDa. Higher molecular-mass species were observed in both samples, suggesting that the chemokine can form oligomers under these conditions.

We subsequently assayed the N-terminally tagged CXCL16 for biological activity. WT CXCR6 transfectants migrated in response to varying concentrations of the chemokine with a typical bell-shaped curve, while naive L1.2 cells were unresponsive (Fig. 4C). Following quantification of the N-terminally tagged CXCL16 protein by ELISA, we compared the chemotactic responses of WT CXCR6 transfectants to those obtained using commercially available CXCL16 (Fig. 4D). Migration to the tagged CXCL16 construct was observed to be an order of magnitude less potent and approximately 60% less efficacious than the commercial recombinant CXCL16, although the N-terminally extended form of CXCL16 was obviously chemotactic, exhibiting the typical bell-shaped dose response.

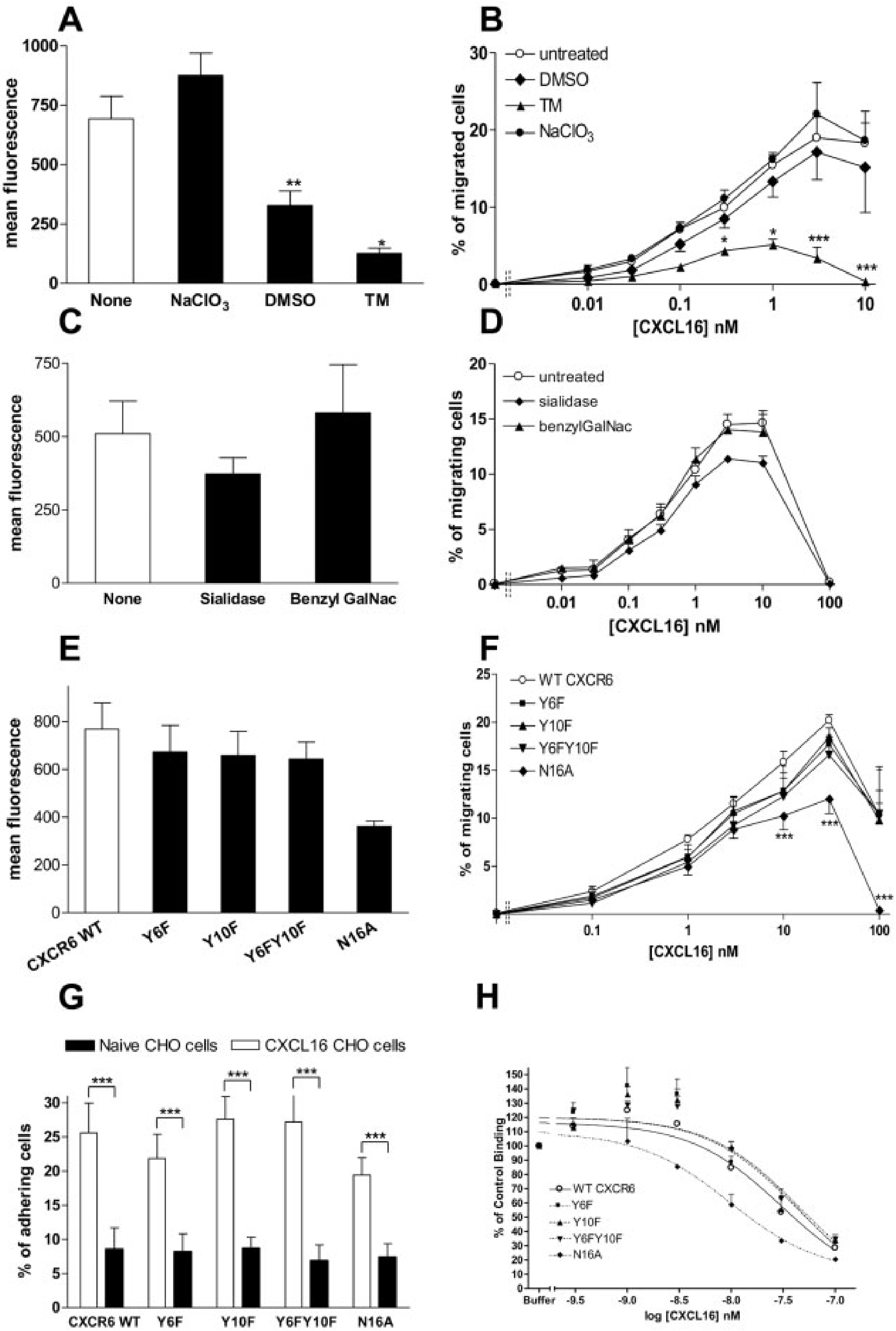

Figure 3. PTM of the CXCR6 N terminus do not play a role in receptor function. (A, B) Cell surface expression levels (A) and chemotactic responses (B) are shown for WT CXCR6 L1.2 transfectants cultured in the presence of 10 μg/mL tunicamycin (TM, in DMSO), DMSO vehicle control, or 10 mM $NaClO_3$. (C, D) Cell surface expression levels (C) and chemotactic responses (D) are shown for WT CXCR6 transfectants incubated for 1.5 h with 0.3 U of sialidase or overnight with 43 mM benzyl-2-acetamido-2-deoxy-α-D-galactopyranoside (BenzylGalNAc). (E, F) Cell surface expression levels (E) and chemotactic responses (F) are shown for WT CXCR6 L1.2 transfectants and those expressing the CXCR6 single-point mutants Y6F and Y10F, N16A and the double CXCR6 mutant Y6FY10F. (G) Relative capacities of the same L1.2 CXCR6 transfectants to adhere to naive CHO cells (black bars) or the CXCL16-CHO-K1 cell line (white bars). (H) Results of competition binding experiments, displacing 125I-CXCL16 from the same L1.2 transfectants with increasing concentrations of unlabelled CXCL16. ***$p<0.001$, **$p<0.01$, *$p<0.05$ using one-way (A, C, E) or two-way (B, D, F, G) ANOVA with Bonferroni's multiple-comparisons test.

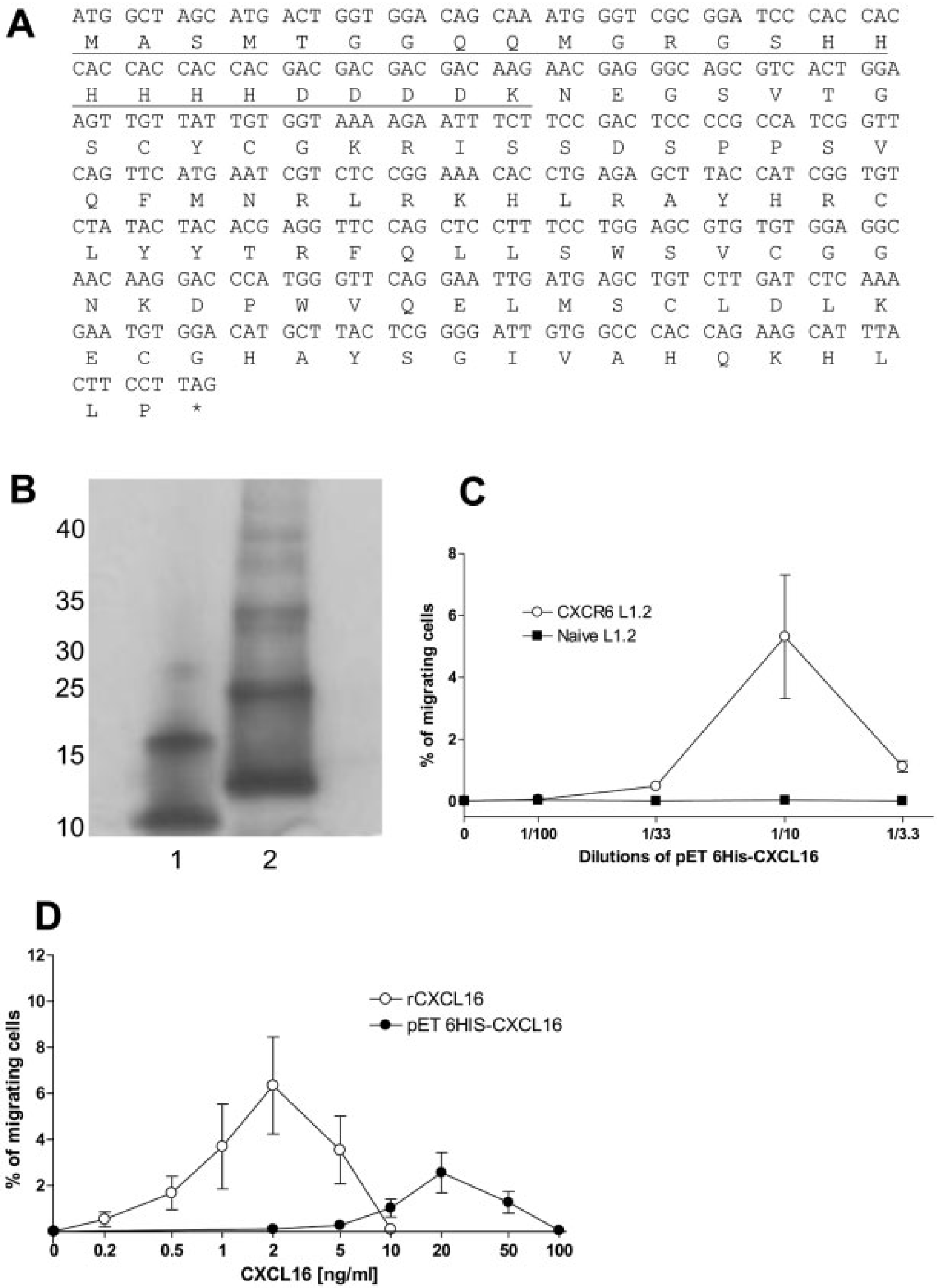

Figure 4. An N-terminally extended form of CXCL16 can function as a CXCR6 ligand. (A) DNA sequence of the pET-6His-CXCL16 construct and the peptide translation. Underlined sequence corresponds to the N-terminal tag consisting of the T7 gene 10 leader peptide, the six histidine residues and the enterokinase cleavage site (DDDDK). (B) Western blot of commercially available recombinant CXCL16 chemokine domain (lane 1) and the pET-6His-CXCL16 construct, following detection with anti-CXCL16 biotinylated antibody and streptavidin-HRP. (C) Chemotactic responses of naive L1.2 cells (filled squares) and WT CXCR6 L1.2 transfectants (open circles) to various dilutions of pET-6His-CXCL16. (D) Chemotactic responses of WT CXCR6 L1.2 transfectants to increasing concentrations of CXCL16 (open circles) or pET-6His-CXCL16 (filled circles).

Acidic residues in ECL2 and TM7 of CXCR6 are critical for binding CXCL16

From the previous data it was concluded that a multi-site model in which the ECL make key contacts with the ligand was more likely to be applicable to the CXCR6:CXCL16 interaction. CXCR6 has five acidic residues located within its three ECL, namely E94, N176, N184, E185, E259 and also a single residue E274 in TM helix 7 (Fig. 2), a highly conserved residue amongst chemokine receptors and postulated to serve as an anchoring point for the ligand [22]. As with the study of the N terminus, these acidic amino acids were changed to their neutral counterpart and for each mutant, cell surface expression was assessed prior to chemotaxis, ^{125}I-CXCL16 binding assays and adhesion to CXCL16-expressing cells.

All mutations within the ECL led to expression at the cell surface comparable to that obtained with the WT receptor, except for the E94Q construct, which had significantly reduced cell surface expression (Fig. 5A). All CXCR6 mutants specifically bound ^{125}I-CXCL16 at levels similar to WT CXCR6, except for the D176N and E274Q mutants, which did not bind the soluble chemokine at all (Fig. 5B). Similarly, all transfectants expressing the mutant constructs responded chemotactically to increasing concentrations of CXCL16 (Fig. 5C), except for the D176N and E274Q constructs, with cells expressing the E274Q mutant being unresponsive, and cells expressing the D176N mutant only responding at the highest concentration of CXCL16 employed, suggestive of a greatly reduced affinity for CXCL16, beyond the limits of detection of our binding assay (Fig. 5B). Cells expressing the E94Q mutant exhibited reduced efficacy in chemotaxis, most likely related to reduced cell surface expression of the construct (Fig. 5C).

When we tested the adhesive properties of these mutant receptors, only the E94Q and D176N mutants showed no significant adhesion to CXCL16-expressing CHO cells (Fig. 5D), correlating with their respective reduced expression and an inability to bind soluble CXCL16. Surprisingly, the E274Q mutant was still able to promote firm adhesion to membrane-bound CXCL16, despite being unable to bind its soluble counterpart (Fig. 5B and D).

We subsequently examined the fate of the two most poorly expressed CXCR6 point mutants, the E3Q and E94Q constructs (Fig. 5E). Intracellular staining revealed that compared to WT CXCR6, a larger proportion of the E3Q mutant receptor remained intracellular, suggesting defects in trafficking of the protein to the cell surface, with the total intracellular and extracellular E3Q staining being less than half that of WT CXCR6 (40.6% $\pm$ 7.9, n=3). As for the E3Q mutant, a larger proportion of the E94Q mutant receptor also remained intracellular (Fig. 5E) with the total intracellular and extracellular CXCR6 staining being around only half that of wild type (51.8% $\pm$ 5.7, n=3).

E274 is critical for the recognition of soluble but not membrane-bound CXCL16

The properties of transfectants expressing the E274Q mutant were subsequently assessed in more detail. Whilst the binding of ^{125}I-CXCL16 to WT CXCR6 was displaced in a dose-dependent manner with increasing concentrations of unlabelled chemokine, the E274Q mutant was unable to specifically bind ^{125}I-CXCL16 (Fig. 6A). In spite of this deficiency, the E274Q construct was still able to facilitate adhesion to cells expressing the membrane-bound form of CXCL16 (Fig. 6B). In keeping with the competition binding data, whilst the adhesion of WT CXCR6 cells to CXCL16-expressing CHO cells was significantly inhibited by the addition of soluble CXCL16, this was not the case for cells expressing the E274Q construct (Fig. 6B). This reveals that CXCR6 can discriminate between membrane-bound and soluble CXCL16 and that E274 is critical in recognising the soluble form of the ligand.

A previous study of CXCR3 has shown that G protein coupling can markedly influence chemokine binding, with the ligand CXCL10 only able to bind to chemokine receptor CXCR3 that is pre-coupled to G protein [23]. In light of these data, the inability of the E274Q mutant to bind CXCL16 might be explained by an inability of the construct to couple to G proteins. We subsequently assessed the effects of G protein uncoupling on binding of CXCL16 by CXCR6. Whilst treatment of WT CXCR6 transfectants with pertussis toxin abolished chemotaxis to CXCL16, it had no discernable effect on the binding of soluble CXCL16 (Fig. 6C), suggesting that G protein coupling does not modulate CXCL16 binding.

Discussion

Both CXCR6 and its ligand CXCL16 have been implicated in the pathogenesis of atherosclerosis by virtue of their expression within the atherosclerotic lesions of humans and apolipoprotein E-deficient mice [8, 24]. Recent data from studies using mice deficient in either CXCR6 or CXCL16 support a role for both ligand and receptor in atherogenesis [25, 26]. Antagonism of the CXCR6:CXCL16 interaction may therefore be of therapeutic potential and a greater understanding of the molecular basis underlying this interaction is highly desirable.

Studies of several chemokine receptors, notably CXCR1, CXCR2, CCR1, CCR2, CCR3 and CCR5, have supported a two-step model of receptor activation [11–13]. In this model, the chemokine ligand is initially tethered to the cell surface by interactions with the receptor N terminus and orientated for subsequent interactions with other regions of the receptor. The tethering event appears to be mediated to a considerable extent by charge, with the receptor N terminus typically rich in acidic side chains, contrasting with the predominantly basic chemokines.

The N terminus of the receptor is also a site for PTM providing additional negative charge such as sulfation and O-linked glycosylation, both of which have been reported to be crucial for ligand binding to the receptors CCR5, CX3CR1 and CXCR3

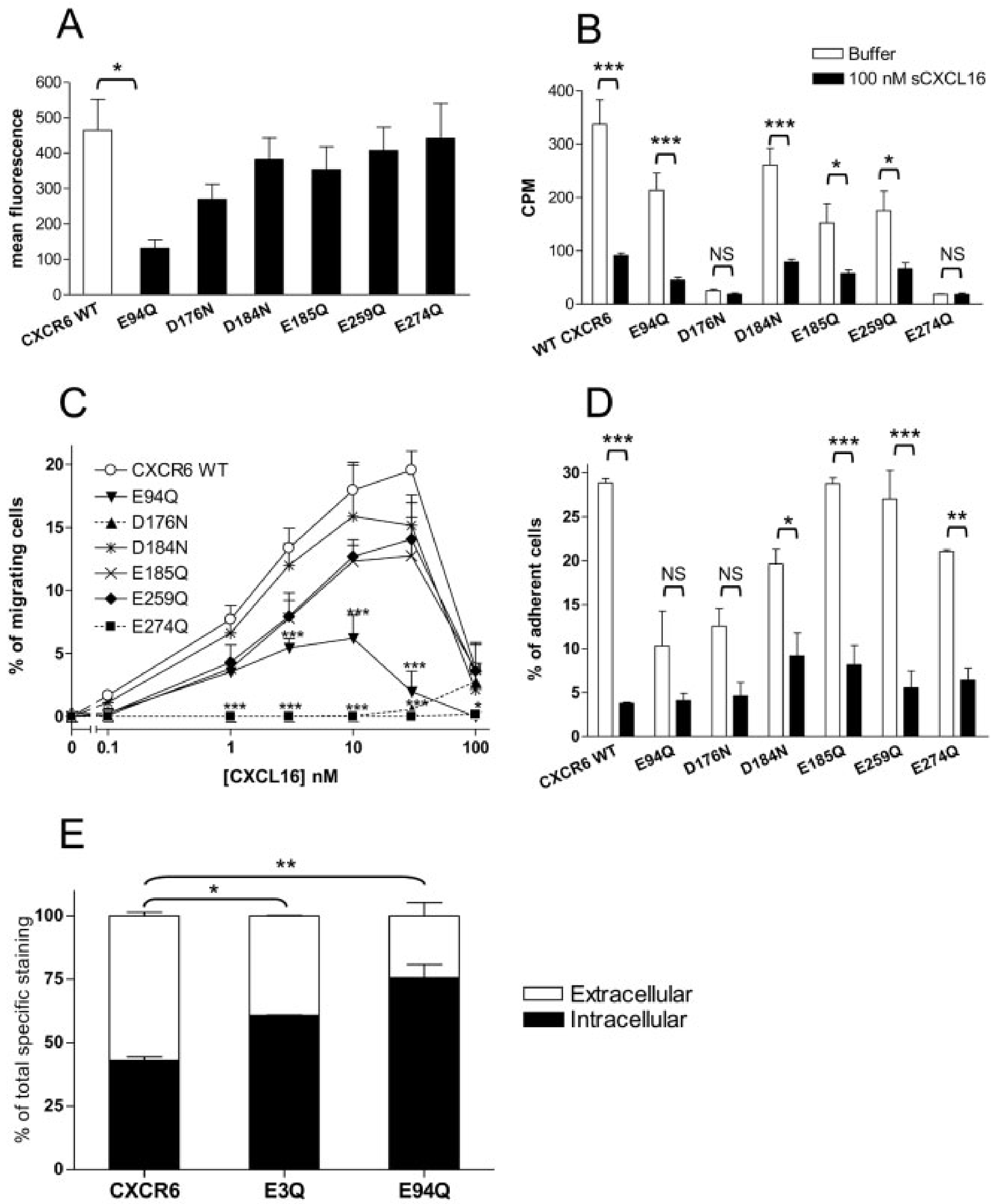

Figure 5. D176 and E274 of CXCR6 are critical for function. (A) Cell surface expression levels of L 1.2 transfectants expressing WT CXCR6 and a variety of extracellular point mutants of CXCR6. (B) Competition binding experiments displacing ^{125}I-CXCL16 from the same CXCR6 transfectants with a 1000-fold excess of unlabeled CXCL16. (C) Chemotactic responses of the same CXCR6 transfectants to increasing concentrations of CXCL16. (D) Adhesion of the same CXCR6 transfectants to naive CHO cells (black bars) or the CXCL16-CHO-K1 cell line (white bars). (E) Relative intracellular and extracellular distribution of WT CXCR6, E3Q and E94Q L1.2 transfectants, expressed as a percentage of total specific staining. $^{***}p<0.001$, $^{**}p<0.01$, $^{*}p<0.05$ using one-way (A, E) or two-way (B–D) ANOVA with Bonferroni's multiple-comparisons test.

[16–19]. The activation event following ligand tethering is postulated to occur in close proximity to the cell membrane and involve the disruption of interactions between side chains of neighbouring TM helices by the chemokine N terminus [13, 14]. Supportive of this postulate, truncation or addition at the chemokine N terminus by one or two amino acids has been shown to produce potent receptor antagonists [21, 27] and recently described small molecule agonists of the chemokine receptors CCR3 and CCR8 have been shown to bind within an intrahelical pocket [28, 29].

Our data described here suggest that current models of chemokine receptor activation are not applicable to the activation of CXCR6 by CXCL16. Although a previous study has highlighted the requirement of R59, R67 and R73 of CXCL16 for high-affinity binding to CXCR6 [15], wholesale mutagenesis of acidic residues within the receptor N terminus, either individually or collectively, was without effect on receptor function. Likewise, inhibition of glycosylation or sulfation, either by mutation or treatment with pharmacological inhibitors, had little effect on CXCR6 function.

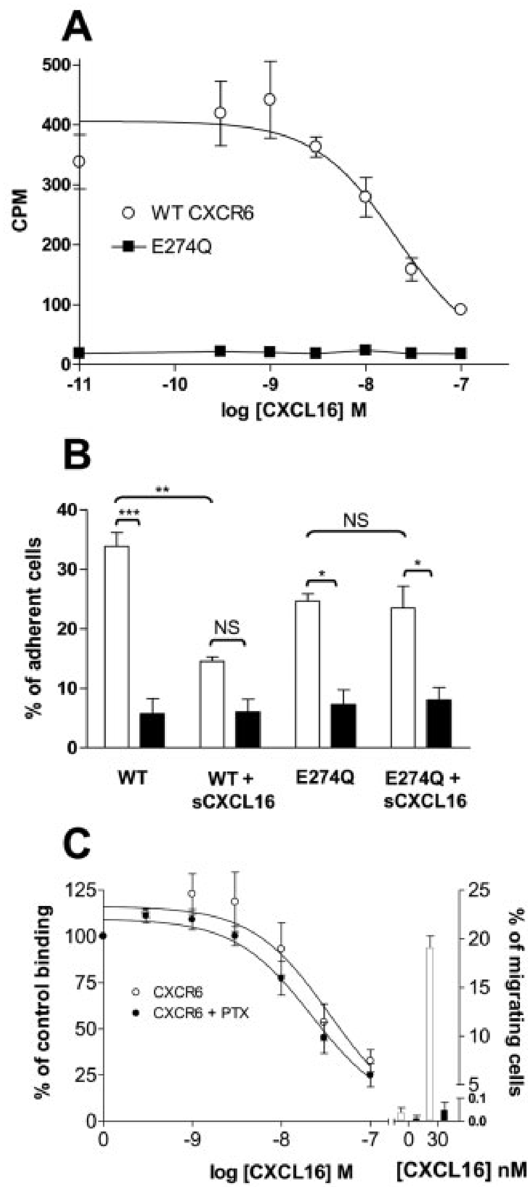

Figure 6. CXCR6 can discriminate between the conformations of membrane-bound and soluble CXCL16. (A) Competition binding experiments displacing ^{125}I-CXCL16 from WT CXCR6 transfectants (open circles) and E274Q CXCR6 transfectants (filled squares) with increasing concentrations of unlabelled CXCL16. (B) Adhesion of WT CXCR6 and E274Q CXCR6 transfectants to naive CHO cells (black bars) or the CXCL16-CHO-K1 cell line (white bars). In blocking experiments, cells were pre-incubated with 1 nM of soluble CXCL16 for 20 min, prior to adhesion. (C) Competition binding experiments displacing ^{125}I-CXCL16 from untreated WT CXCR6 transfectants (open circles) or WT CXCR6 transfectants following overnight incubation with pertussis toxin (PTX, filled circles). The right-hand axis shows the chemotactic responses of the same untreated CXCR6 transfectants (white bars) and pertussis toxin-treated CXCR6 transfectants (black bars) to CXCL16. $^{***}p<0.001$, $^{**}p<0.01$, $^{*}p<0.05$ using two-way ANOVA with Bonferroni's multiple-comparisons test.

Of note was the finding that the loss of a negative charge at position 3 in the CXCR6 sequence (E3Q mutation) resulted in a dramatic loss of cell surface expression. When both intracellular and extracellular immunoreactivity was assessed and compared to WT CXCR6, less than half the total expression of WT CXCR6 was observed for the E3Q mutant, with proportionally less of the mutant construct trafficking to the surface, the remainder residing in a large intracellular pool.

This may explain the previous finding of Duggal and colleagues [30], regarding a single nucleotide polymorphism of CXCR6, G1469A, which results in the naturally occurring E3K mutant of CXCR6. Individuals homozygous or heterozygous for the WT allele were found to be significantly more likely to die a *Pneumocystis carinii* pneumonia-mediated AIDS-related death than those homozygous for the CXCR6 G1469A allele. Assuming that the CXCR6 E3K mutant struggles to traffic to the cell surface, just as the E3Q mutant we describe here, it can be hypothesised that HIV-1 utilises CXCR6 as a co-receptor for viral entry during the later stages of infection and that reduced expression of CXCR6 conferred by the CXCR6 G1469A allele is protective. Such a scenario is plausible, as CXCR6 has previously been shown to facilitate the entry of both macrophage- and T cell line-tropic strains of HIV-1 [31].

Mutation of acidic residues located outside of the CXCR6 N terminus was revealing. Replacement of D176 (ECL2) and E274 (TM7) led to a dramatic loss of receptor function as determined by chemotaxis, despite excellent levels of cell surface expression. This loss of function in both mutants could be attributed to a loss in the ability to bind soluble CXCL16 as deduced by binding assays. A glutamate residue at position 6 of TM7, as typified by E274 of CXCR6, is highly conserved within chemokine receptors [22] and has been postulated to act as a point of contact for agonists of chemokine receptors [28, 29].

Curiously, despite being unable to bind soluble CXCL16, cells expressing the E274Q mutant of CXCR6 were able to robustly support adhesion, suggesting that the mutant receptor adopts a conformation able to interact with the membrane-bound but not the soluble form of CXCL16. This also suggests that rather than simply acting to tether the chemokine to the cell membrane, the mucin stalk also infers a particular conformation upon the chemokine domain of CXCL16, making it subtly distinct from that of the soluble chemokine. Previous studies have shown that endogenous shedding of the membrane-bound forms of CXCL16 and the related chemokine CX3CL1 is carried out by the actions of the metalloproteinase ADAM10 [32, 33]. Although the exact site of cleavage has not been defined at the primary sequence level, SDS-PAGE analysis suggests that cleavage is likely to take place at the membrane interface, shedding CXCL16 into the supernatant with a significant proportion of the mucin stalk still attached [33].

Our finding that although functional in chemotaxis assays, the stalk-bound CXCL16 was unable to displace soluble CXCL16 from CXCR6 in binding assays, suggests that these two forms of CXCL16 are allotopic, likely stabilising distinct CXCR6 conformations

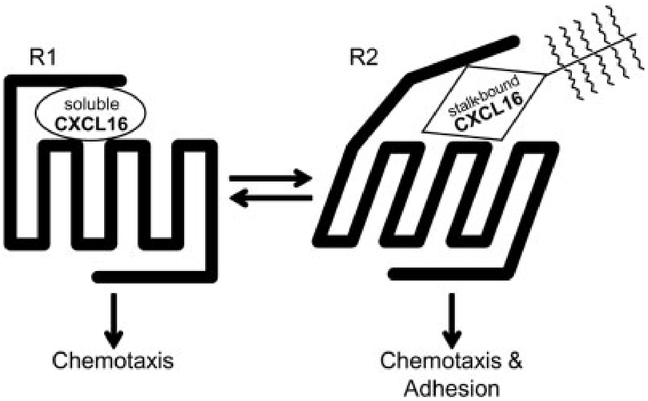

Figure 7. A cartoon highlighting the interactions of soluble and stalk-bound CXCL16 with CXCR6. Soluble CXCL16 (oval) is shown to have a distinct conformation to that of stalk-bound CXCL16 (diamond). Both ligands stabilise distinct receptor conformations (R1 and R2), giving rise to the functions associated with CXCR6 activation.

(Fig. 7). This is in contrast to a study of CX3CR1 in which stalk-bound CX3CL1 was shown to effectively displace soluble CX3CL1 when the two ligands were added simultaneously to the cells, as is the case with the binding assay we describe here [34]. The ability of soluble CXCL16 to antagonise CXCL16-mediated adhesion is therefore likely to be a result of stabilising a CXCR6 conformation which cannot recognise the membrane-anchored form (R1). Since we have previously shown that soluble CXCL16 can also induce the rapid endocytosis of cell surface CXCR6, this process may also play a part in the antagonism of CXCL16-mediated adhesion [35].

Our observation that the E274Q CXCR6 mutant could promote adhesion but not signalling, tallies with results of a previous study, in which pertussis toxin treatment of CXCR6 transfectants was found to ablate chemotactic responses to soluble CXCL16 but not adhesion to the membrane-bound form of CXCL16, highlighting that signalling and adhesion are independent functions of CXCR6 [10]. This is also true for CX3CL1-mediated adhesion [36, 37]. Interestingly, an N-terminally extended form of CXCL16 was still able to induce significant CXCR6 activation, suggesting that a model in which the chemokine N terminus interacts with an intrahelical binding pocket containing E274 of CXCR6 is unlikely to explain activation of this chemokine receptor. More likely is a scenario whereby binding of CXCL16 to the ECL of CXCR6, involving a critical interaction with D176 of ECL2, promotes a conformational change in CXCR6 needed for G protein recruitment and activation. This is reminiscent of an earlier report from our group in which ECL2 of CXCR3 was shown to be critical for ligand binding and activation [38].

In summary, the data we describe here dissecting the activation of CXCR6 by ligand do not support the typical model for receptor activation in which the N terminus of the chemokine activates an intrahelical pocket containing a conserved glutamate residue. This may have implications for the identification of small-molecule antagonists of CXCR6, which typically occupy such binding sites. Since the mutation of E274 resulted in a complete loss of soluble CXCL16 binding, it may be that the interaction of this residue with a neighbouring TM side chain is crucial for a correct CXCR6 conformation, and that small molecules which disrupt such an interaction will prove useful antagonists of CXCR6 activation. In contrast, the dissection of the adhesive and chemotactic functions of the receptor suggests that it may be difficult to generate small-molecule antagonists of CXCR6 which inhibit both the chemotactic and adhesive properties of the receptor.

Materials and methods

Materials

Reagents were purchased from Sigma-Aldrich (Poole, UK) and Invitrogen (Paisley, UK), unless stated otherwise. Chemokines were from PeproTech EC Ltd. (London, UK) or R&D Systems (Abingdon, UK). The mouse anti-HA antibody was from Covance Research Products (Berkeley, CA), and IgG1 isotype control antibody was from Sigma-Aldrich. FITC-conjugated goat anti-mouse antibody was from DAKO Cytomation (Ely, UK). The HA-tagged clone of CXCR6 was from UMR cDNA Resource Center (www.cdna.org) and the CXCL16 clone IRATp970B0959D6 was from the German Resource Center for Genome Research (Berlin, Germany). Bugbuster was from Novagen (Darmstadt, Germany) and TALON spin columns from Clontech Laboratories (Mountain View, CA).

Mutagenesis of HA-CXCR6

Point mutants of a cDNA encoding CXCR6 were generated by PCR using the QuikChange^{TM} site-directed mutagenesis kit (Stratagene, The Netherlands). The authenticity of the resulting constructs was confirmed by double-stranded DNA sequencing (MWG Biotech, UK).

Cell culture and transfections

CHO cells were maintained in F12-HAM media, with 10% FBS, 1 mM L-glutamine, 1 U/mL penicillin and 1 μg/mL streptomycin, and detached using trypsin. L1.2 cells were maintained in HEPES-modified RPMI 1640 medium, with 10% FBS, 100 U/mL penicillin, 0.1 mg/mL streptomycin, 2 mM L-glutamine, 1 mM sodium pyruvate, 50 μM β-mercaptoethanol and 1X MEM non-essential amino acids. All cells were kept at 37°C, 5% CO_2. CHO cells were stably transfected with full-length CXCL16 cloned into pCDNA3 vector, using the FuGENE6 kit from Roche Diagnostics (Basel, Switzerland). G418 (1 mg/mL) was used for positive selection. L1.2 cells were transiently transfected by electroporation with 1 μg DNA/1×10^6 cells, as described previously [39], and incubated overnight with 10 mM sodium butyrate.

Flow cytometry

Extracellular staining was carried out using an anti-HA antibody or isotype control followed by a FITC-conjugated secondary antibody as previously described [28]. For intracellular staining, cells were fixed with 2% paraformaldehyde and staining was performed in the presence of 0.5% saponin. Samples were analysed on a FACSCalibur flow cytometer (Becton Dickinson, Mountain View, CA).

Chemotaxis assay

The chemotactic responsiveness of cells was ascertained using ChemoTx^{TM} plates (Receptor Technologies Ltd., Oxon, UK), as described previously [39]. Cells were allowed to migrate for 5 h in a humidified chamber at 37°C, 5% CO_2, after which the migrating cells traversing the filter were counted using a haemocytometer. Migrating cells were expressed as a percentage of the total cells applied to the filter. EC_{50} values for chemotaxis were generated following fitting of the data with PRISM version 4.03 software (GraphPad, San Diego, CA).

Competitive radioligand binding assay

Whole-cell binding assays on L1.2 cells transiently expressing human CXCR6 were performed as described previously [39], using 0.1 nM ^{125}I-CXCL16 (PerkinElmer Life and Analytical Sciences, Waltham, MA) and various concentrations of unlabelled chemokine. In some assays, supernatant from naive CHO-K1 cells or CXCL16-CHO-K1 cells was used as a competitor in increasing amounts up to 90% of the total reaction volume. A Canberra Packard Cobra 5010 γ-counter (Canberra Packard, UK) was used to assess the level of competition for binding of ^{125}I-CXCL16 with the receptor.

Adhesion assay

Naive and CXCL16-expressing CHO stable cells were seeded overnight at 3×10^4 cells in 200 μL of culture medium per well in a 96-well plate. The next day, naive and transiently transfected L1.2 cells were labelled with 2 μM calcein-acetoxymethylester in RPMI (37°C for 30 min), then washed three times with RPMI and resuspended at 1.2×10^5 cells/200 μL in RPMI with 1% BSA. Where mentioned, L1.2-CXCR6 cells were pre-incubated with soluble CXCL16 for 20 min. L1.2 transfectants or naive L1.2 cells (200 μL) were plated onto the seeded CHO cells, centrifuged (500 rpm, 1 min) and incubated (37°C, 5% CO_2) for 1 h. Fluorescence was quantified before and after washing (with RPMI by gentle pipetting) using a Fluostar Galaxy plate reader from BMG Labtechnologies (Offenburg, Germany) at 485 nm excitation and 520 nm emission.

His-CXCL16 generation and analysis by western blot

The chemokine domain of CXCL16 was amplified by PCR using oligonucleotide primers that incorporate a 6-histidine tag and an enterokinase cleavage site upstream of the deduced mature CXCL16 sequence. Following amplification and digestion, the product was ligated into pET11 vector (Stratagene, CA). Expression in the *E. coli* strain BL21 (DE3) was induced by isopropyl-thio-2-D-galactopyranoside. Cell pellets were harvested and lysed with Bugbuster, and the His-tagged protein purified by TALON spin columns (Clontech). Purified proteins were separated on tricine 10–20% gels, transferred to nitrocellulose membrane from Invitrogen and probed with biotin-coupled anti-CXCL16 and streptavidin-HRP. Visualisation was performed with enhanced chemiluminescence from GE Healthcare (UK).

Data and statistical analysis

Data are representative of at least three experiments and are expressed as the mean ± SEM. Data were analysed with a relevant statistical test where stated, using PRISM version 4.03 software.

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Abbreviations: **ECL:** extracellular loop · **PTM:** post-translational modification · **TM:** transmembrane

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